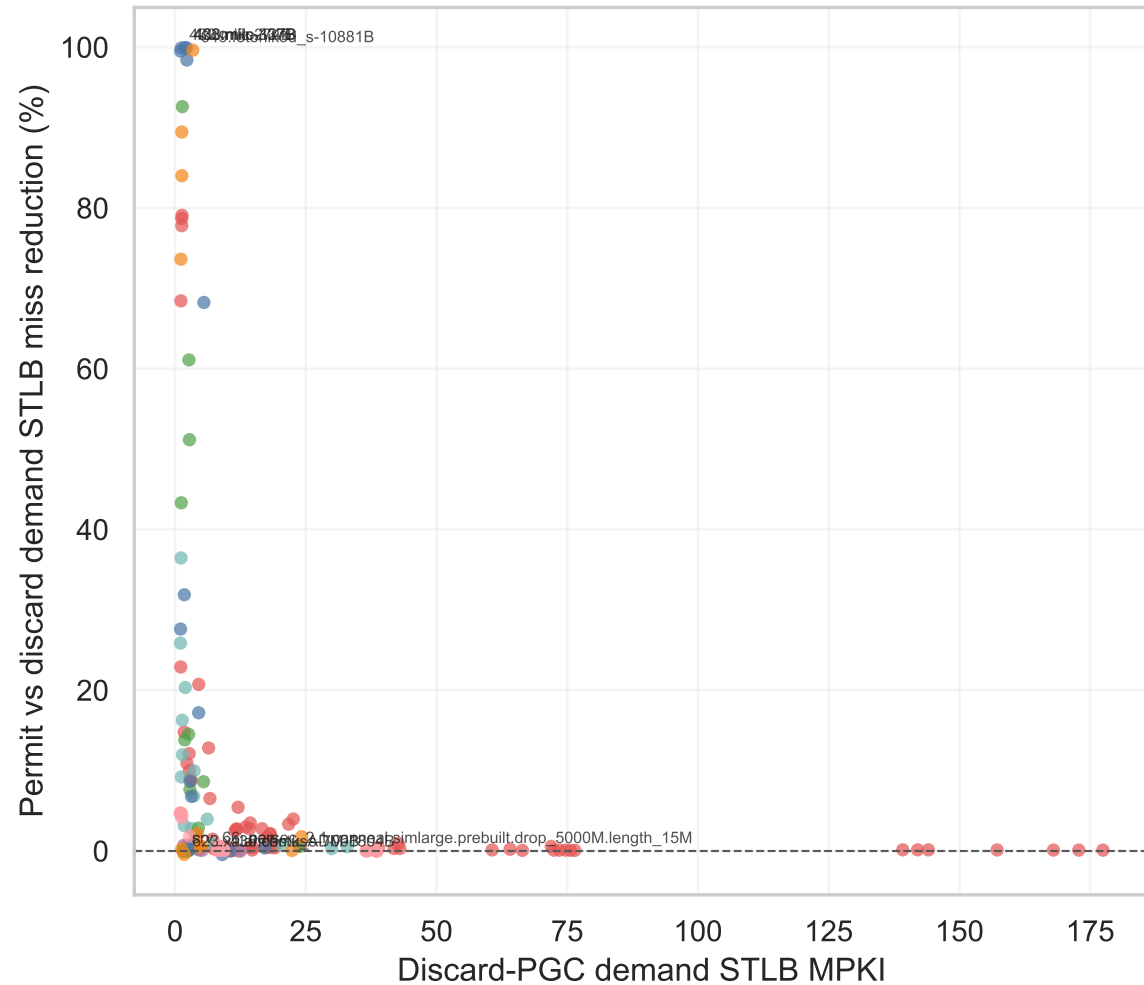
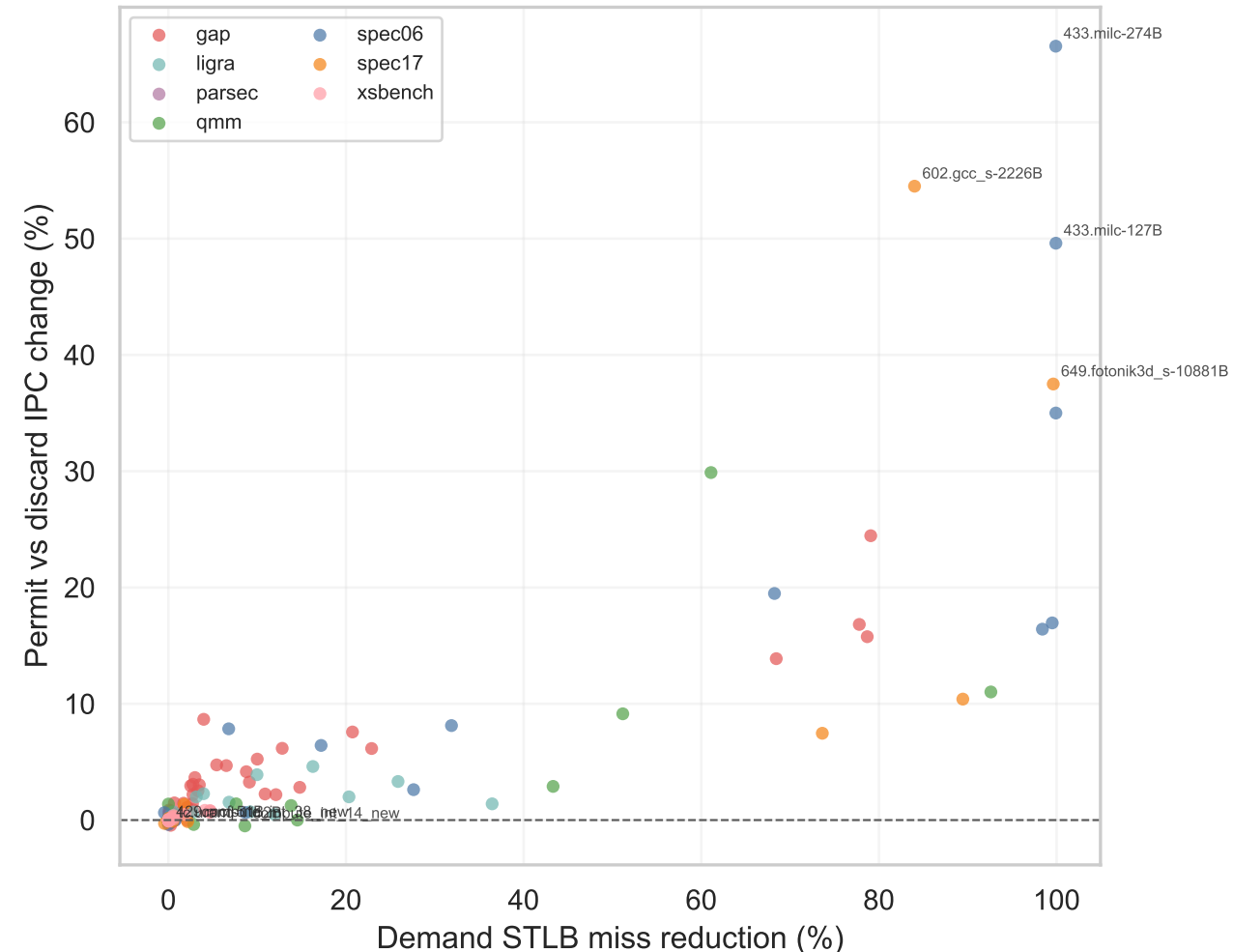


01. Demand-side opportunity and observed outcome

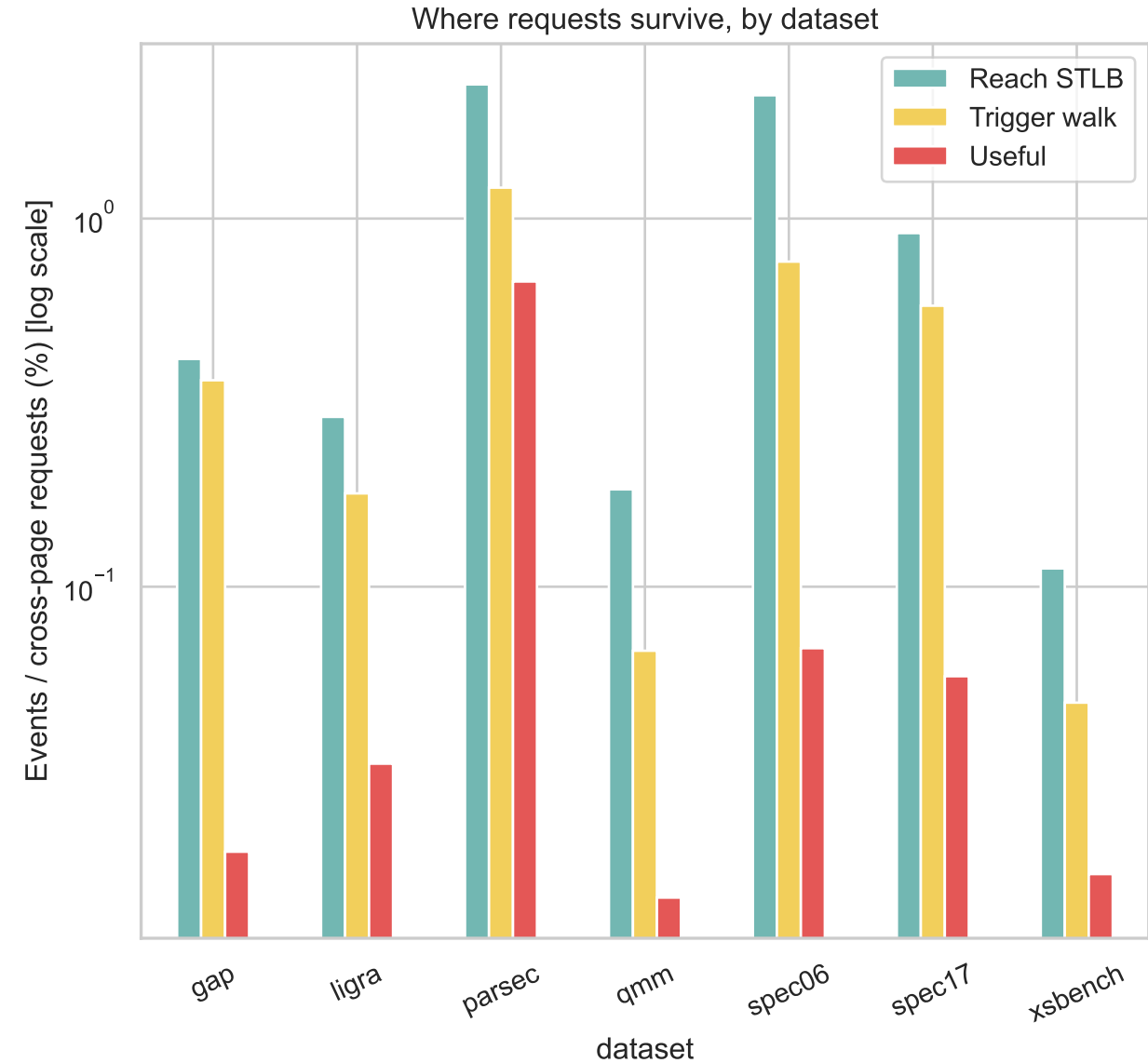
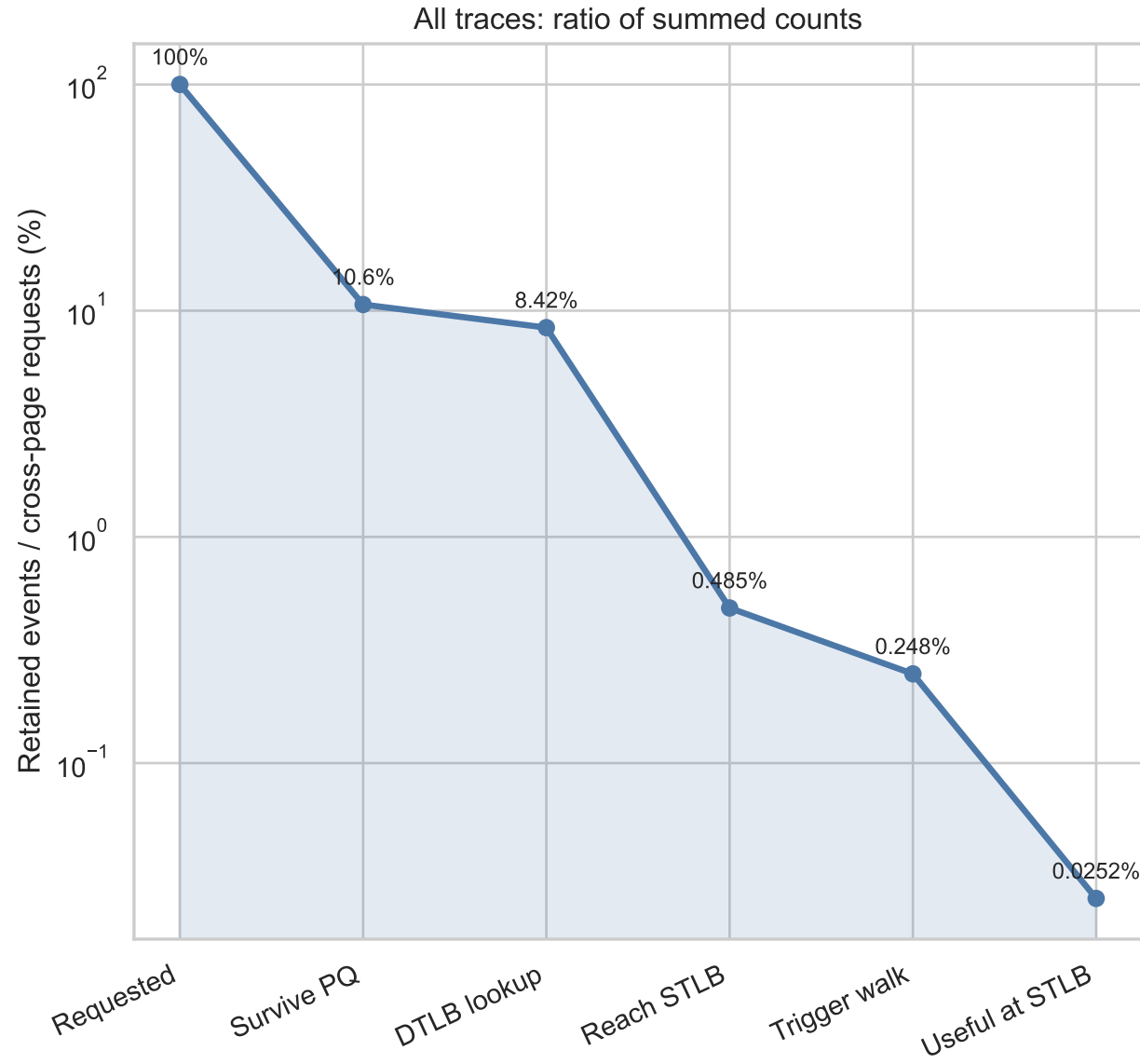
Opportunity does not guarantee coverage



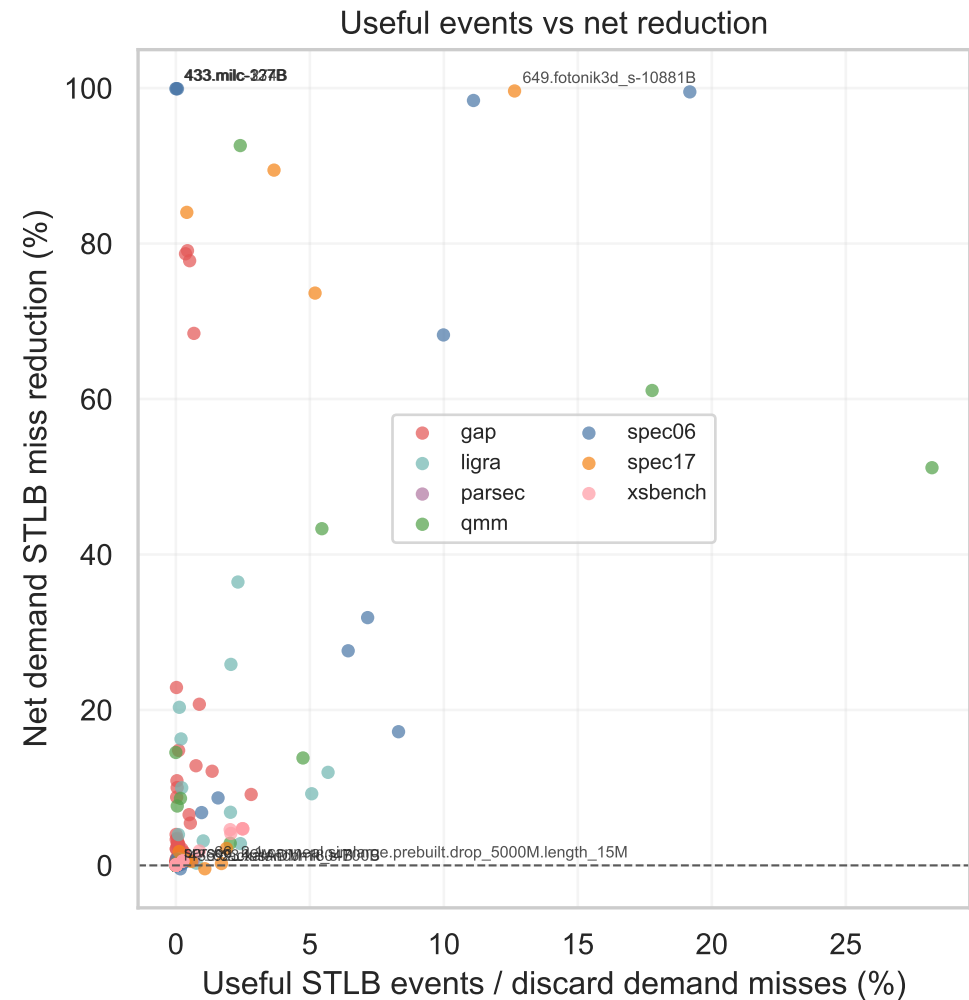
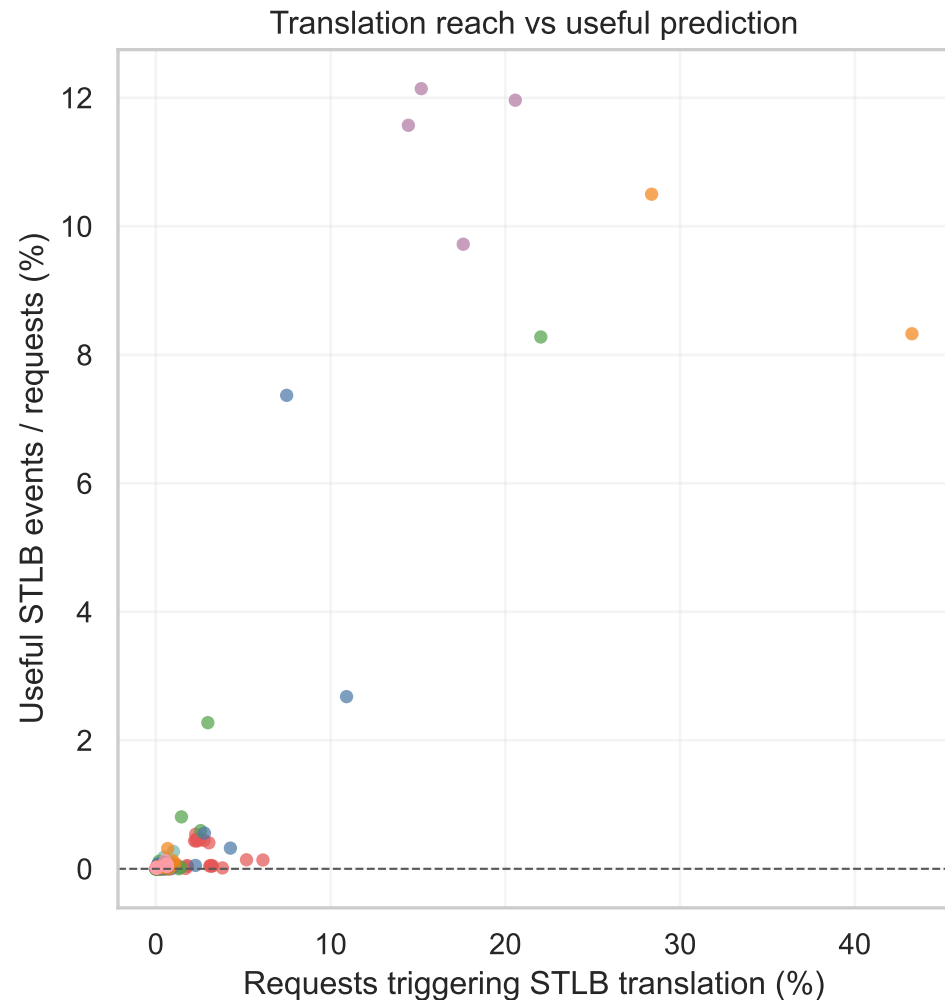
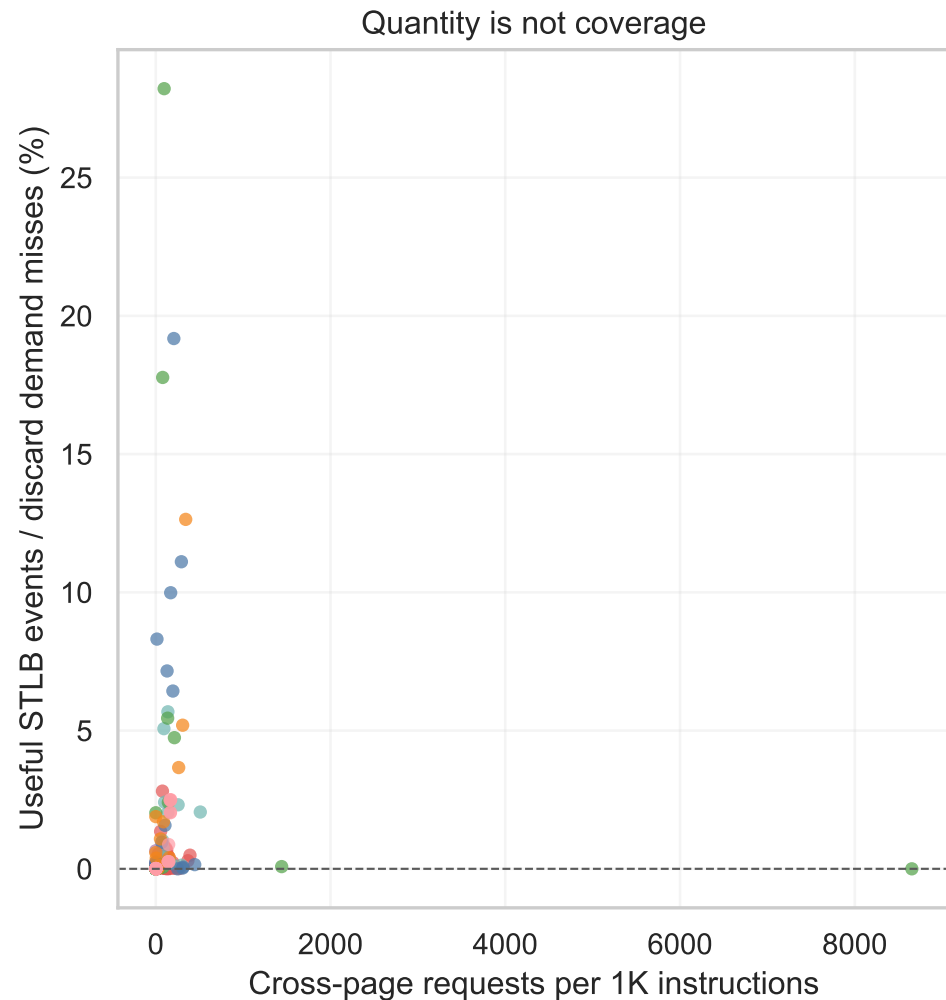
TLB result vs mixed-system IPC result



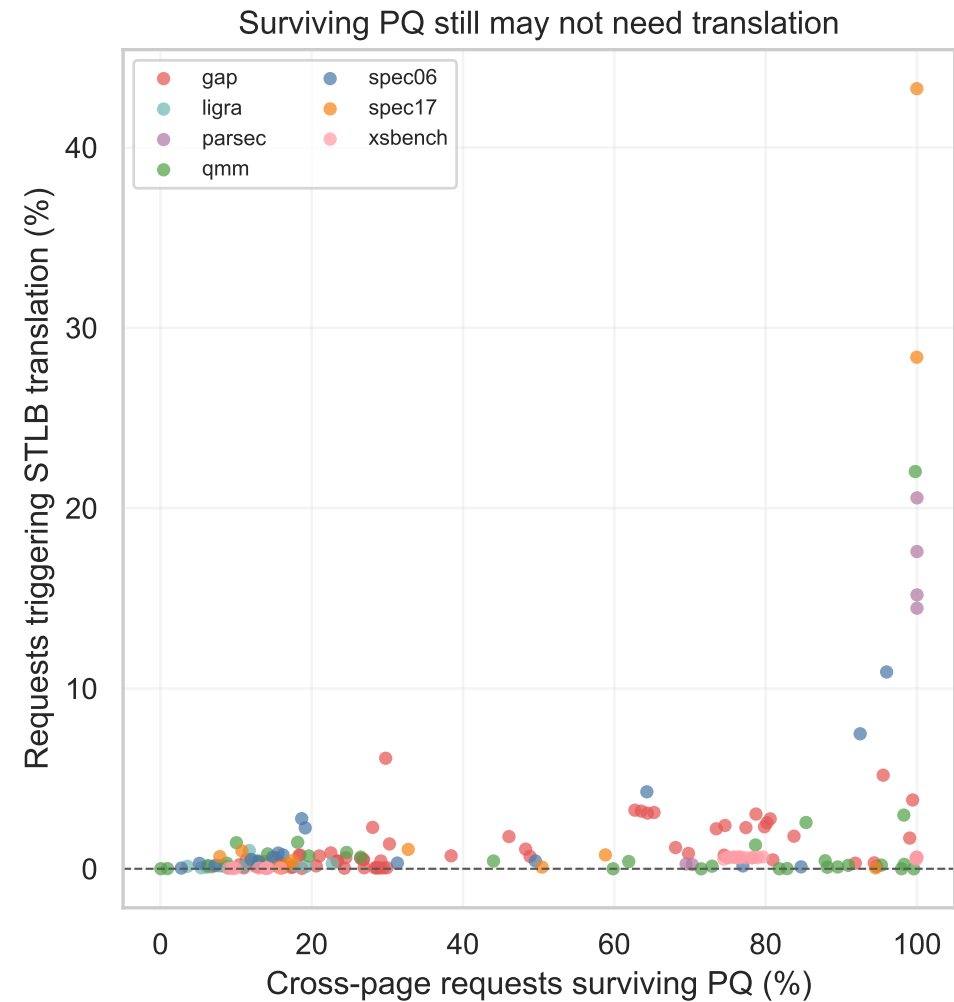
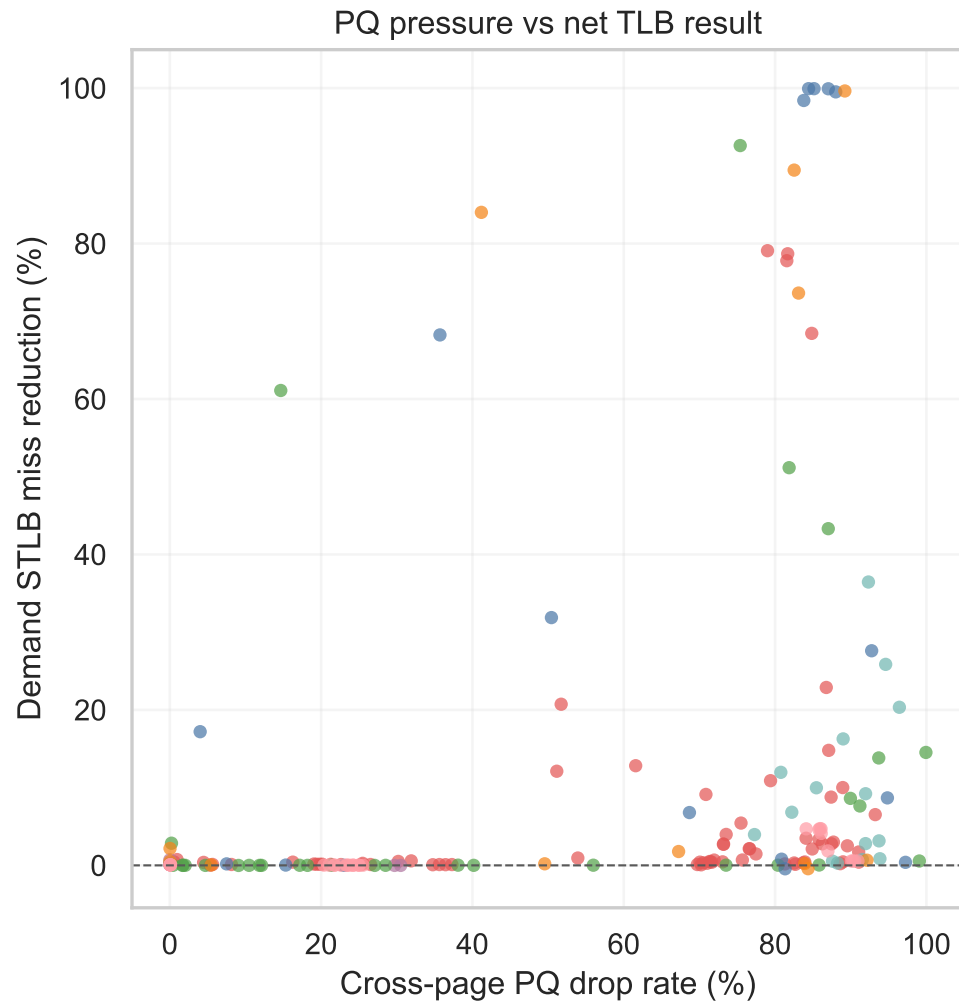
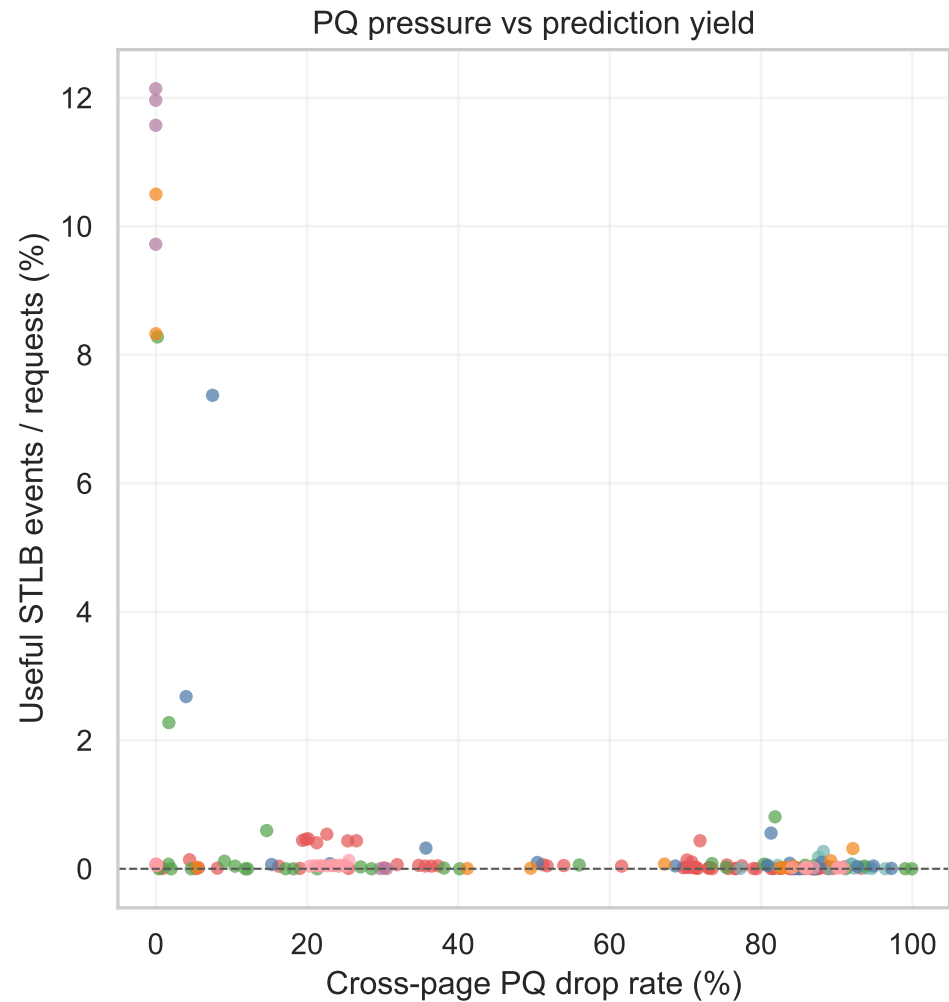
02. Cross-page translation evidence funnel



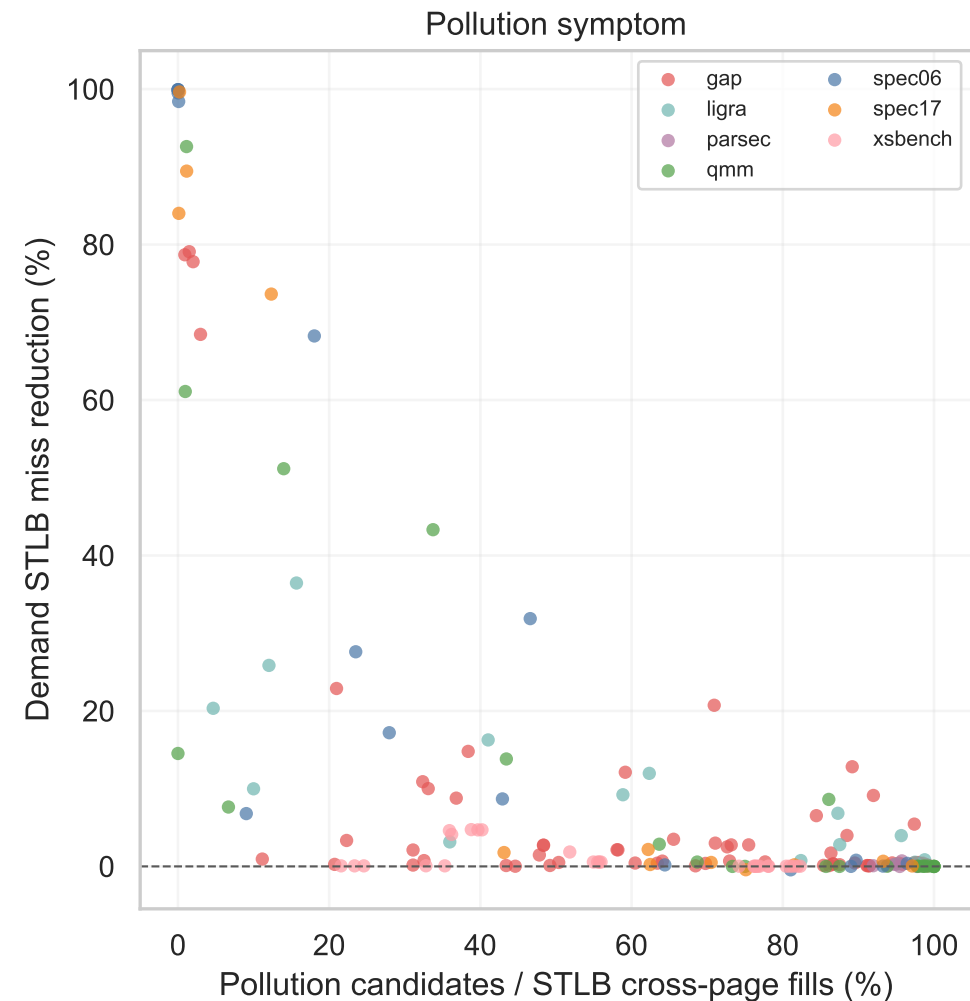
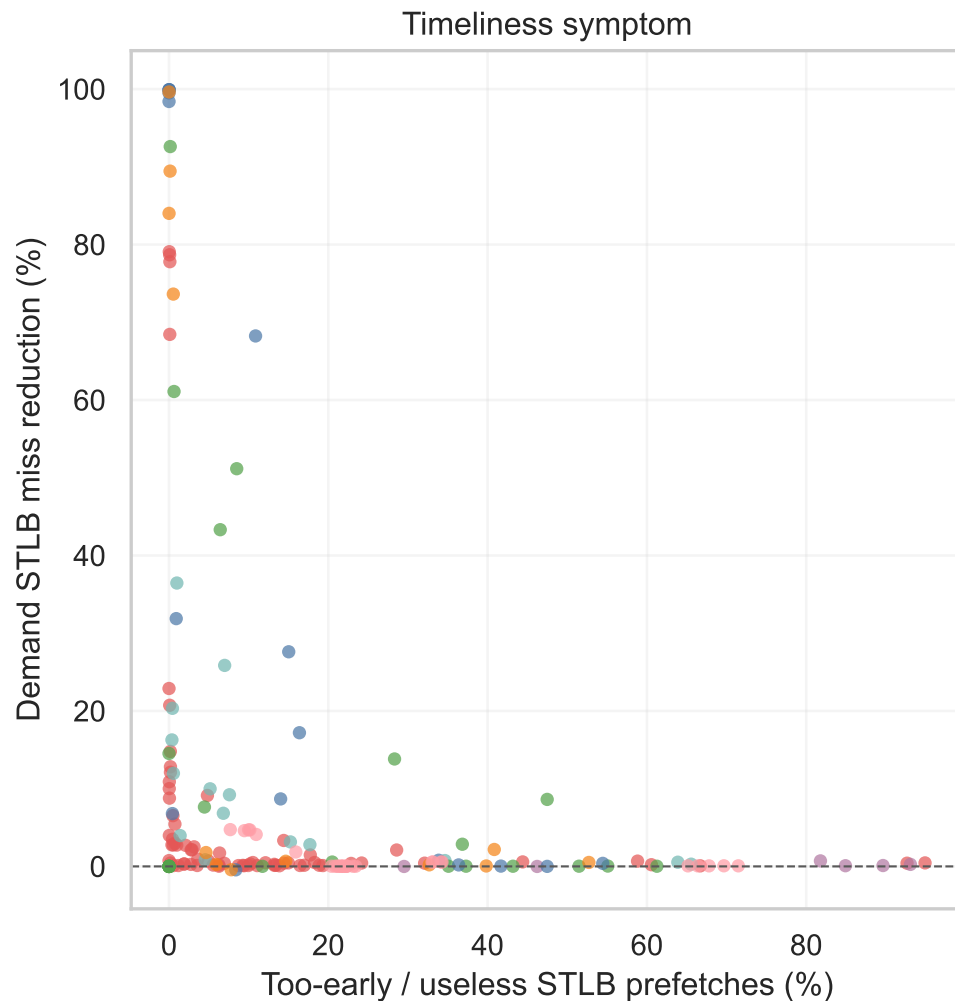
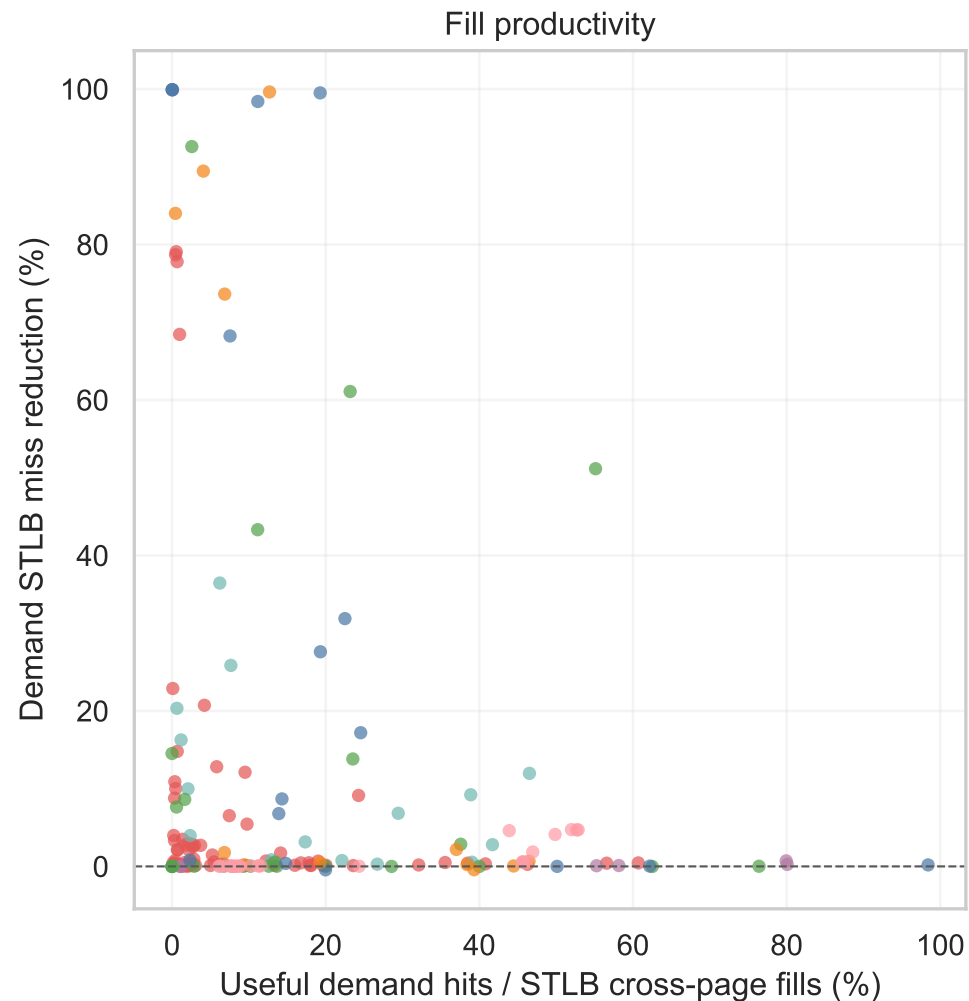
03. Does vBerti generate the right cross-page translations?



04. Is PQ dropping the main bottleneck?

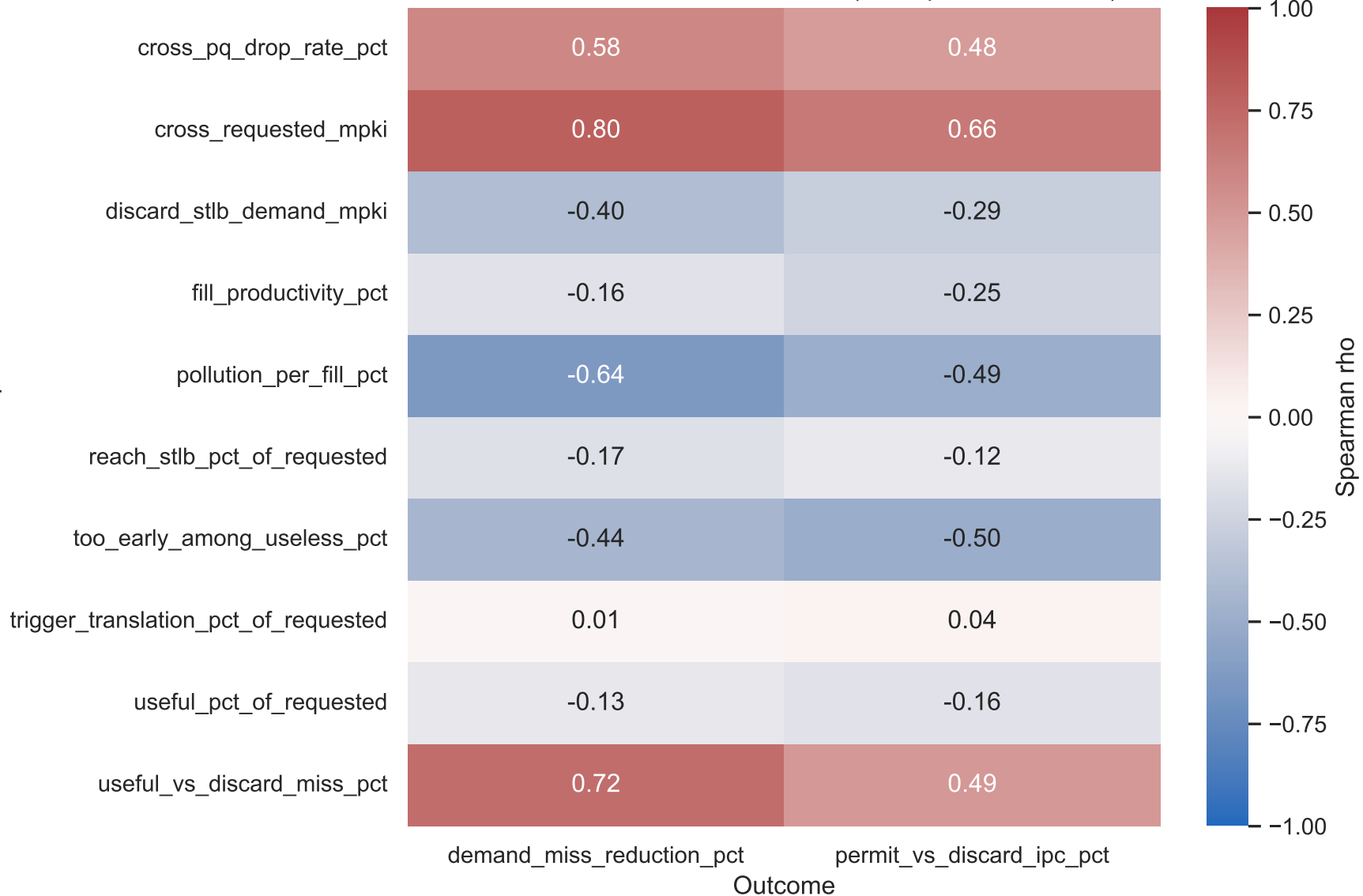


05. Downstream timeliness and pollution checks

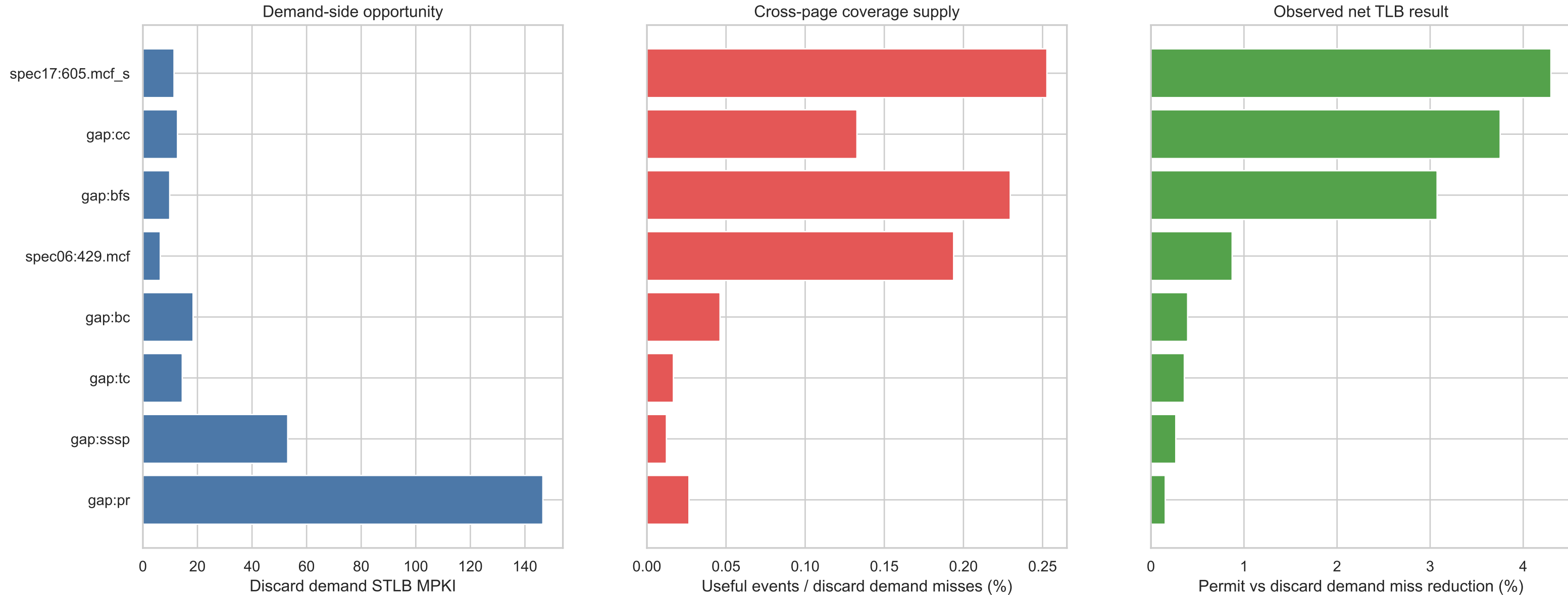


Trace-level monotonic associations (descriptive, not causal)

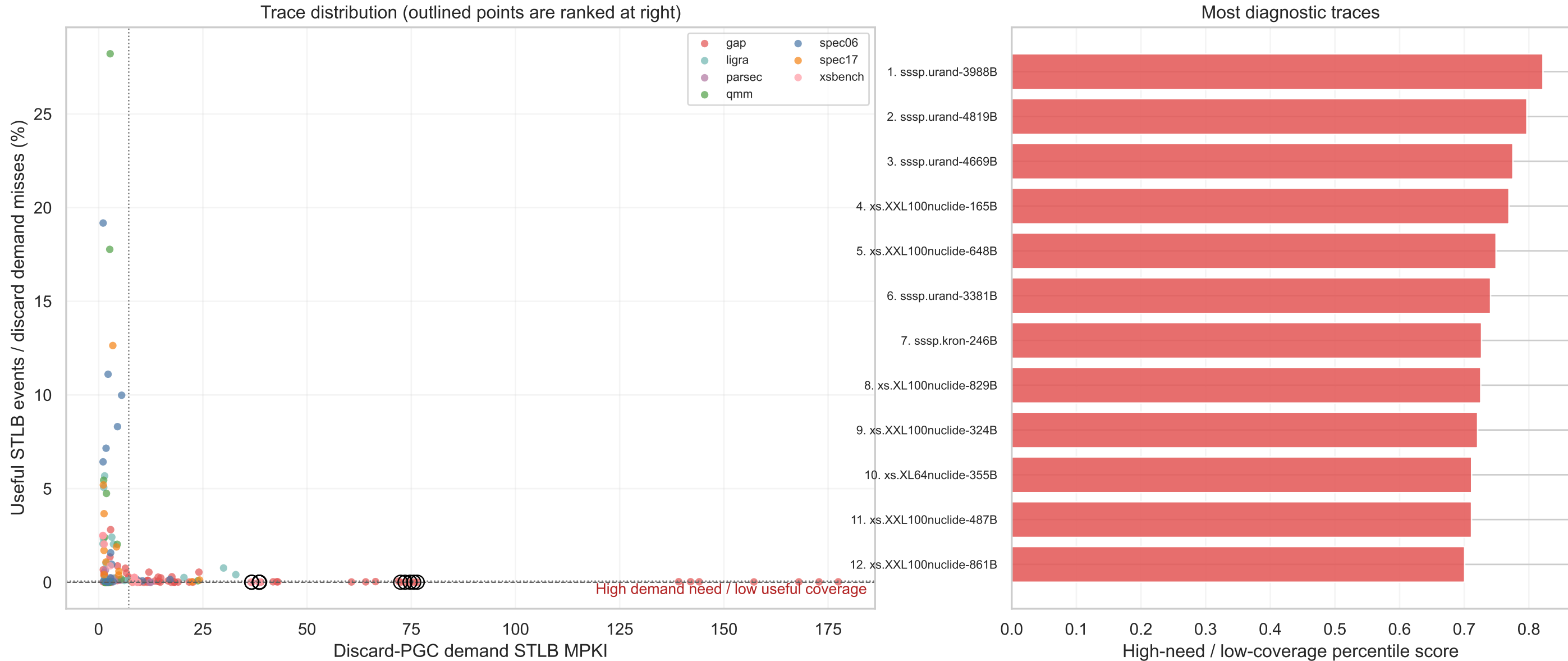
Candidate explanation



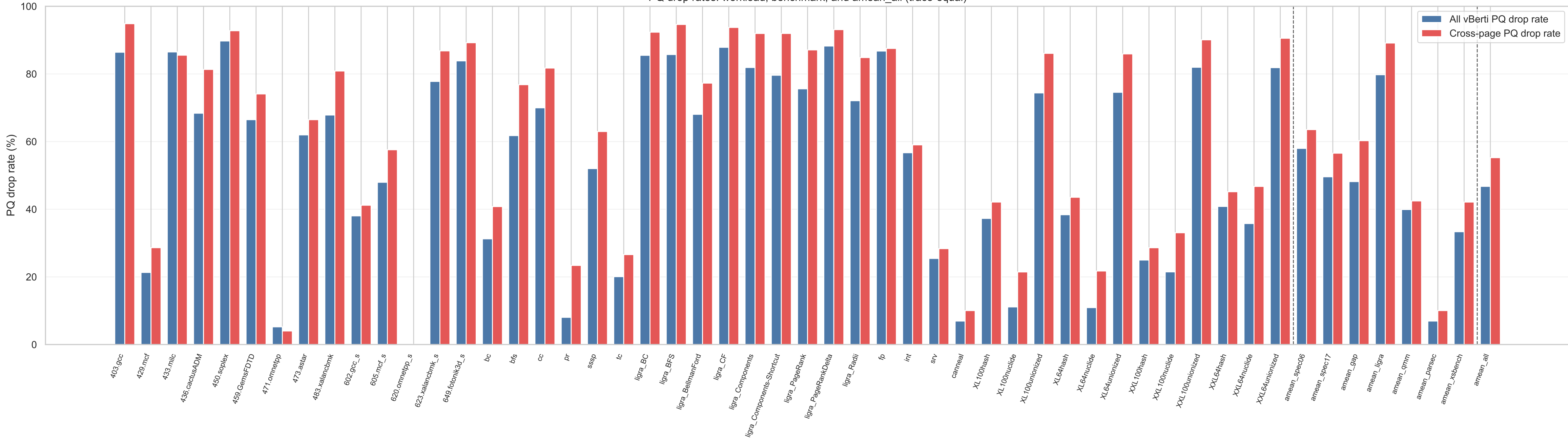
07. GAP and mcf: demand need vs useful cross-page coverage



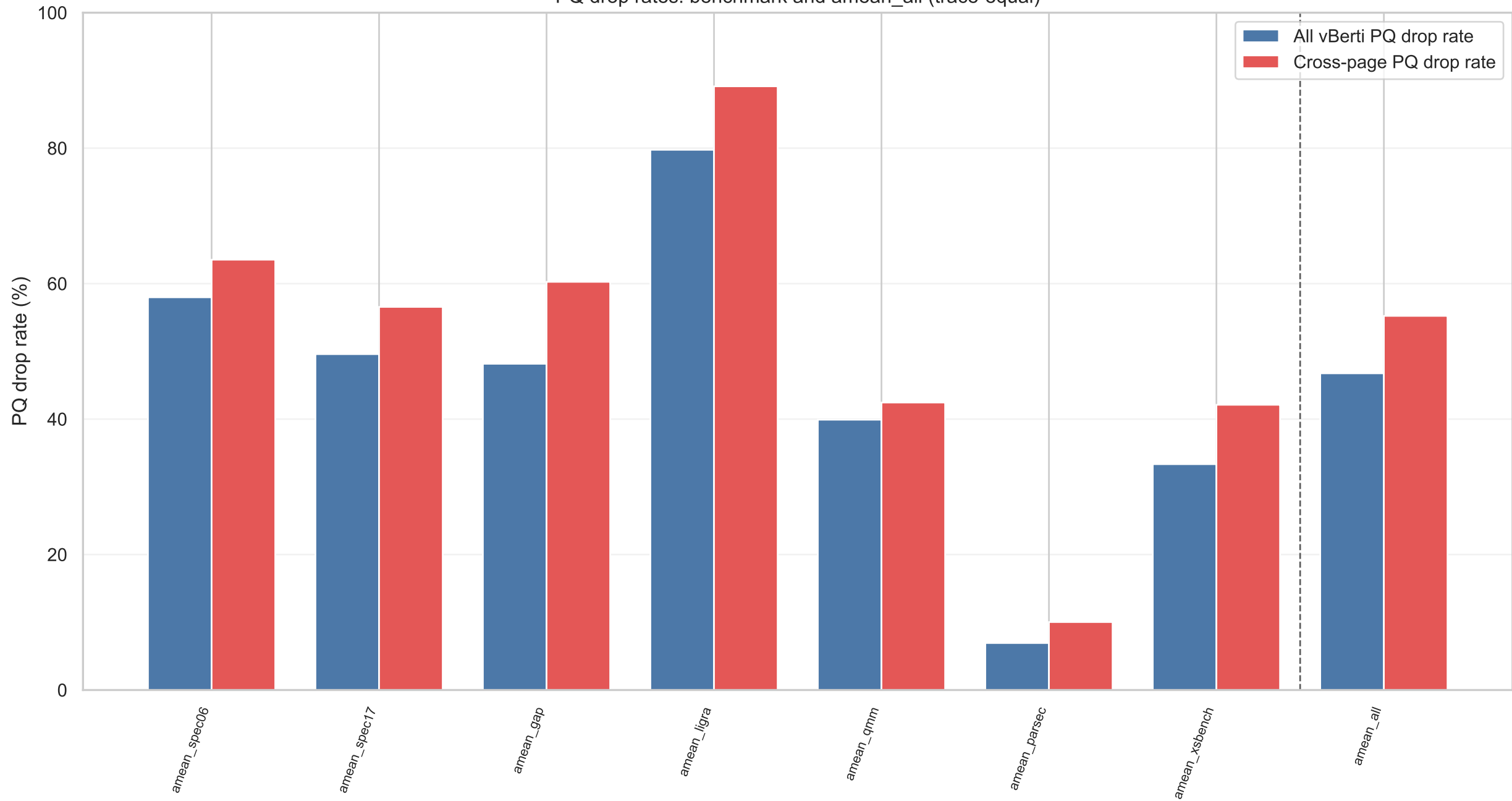
08. Direct test of the high-need / low-coverage hypothesis



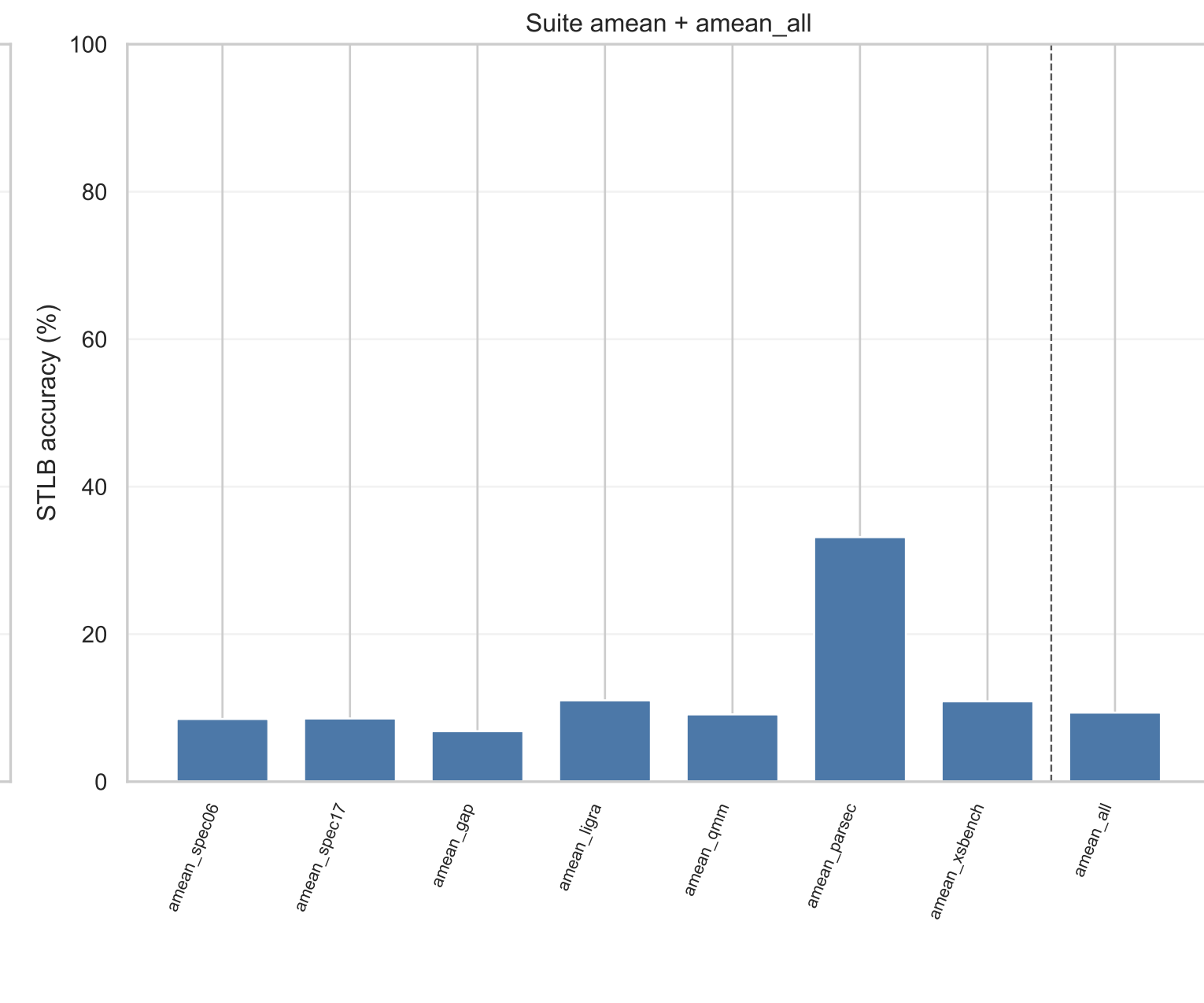
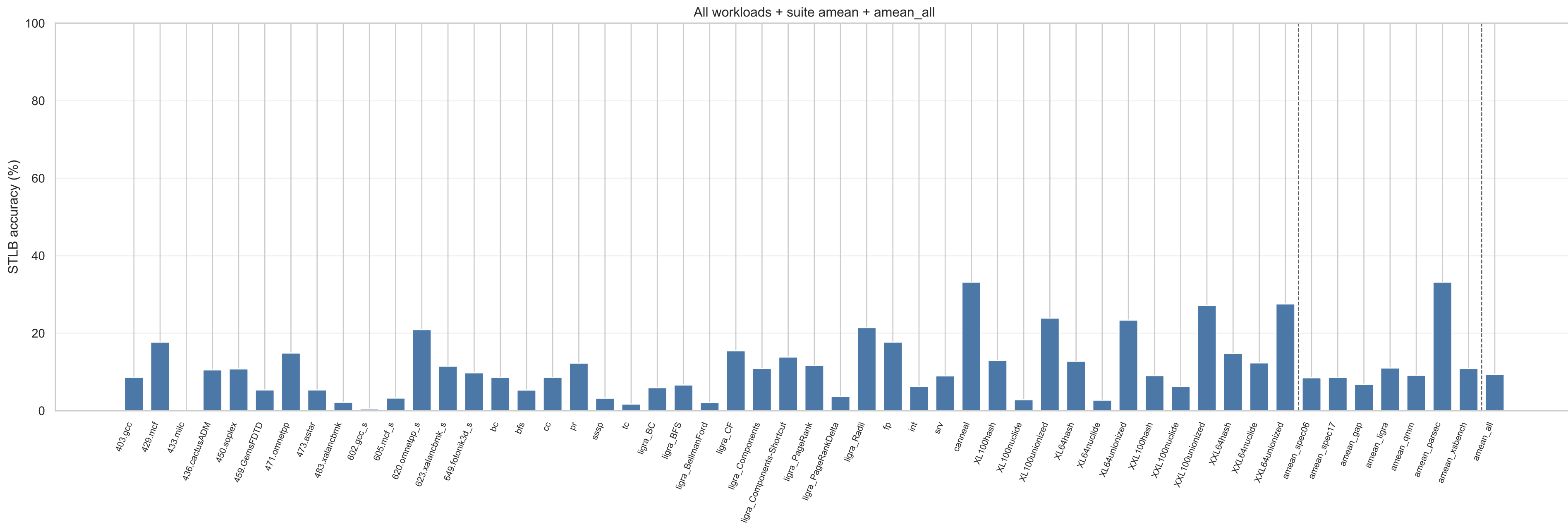
PQ drop rates: workload, benchmark, and amean_all (trace-equal)



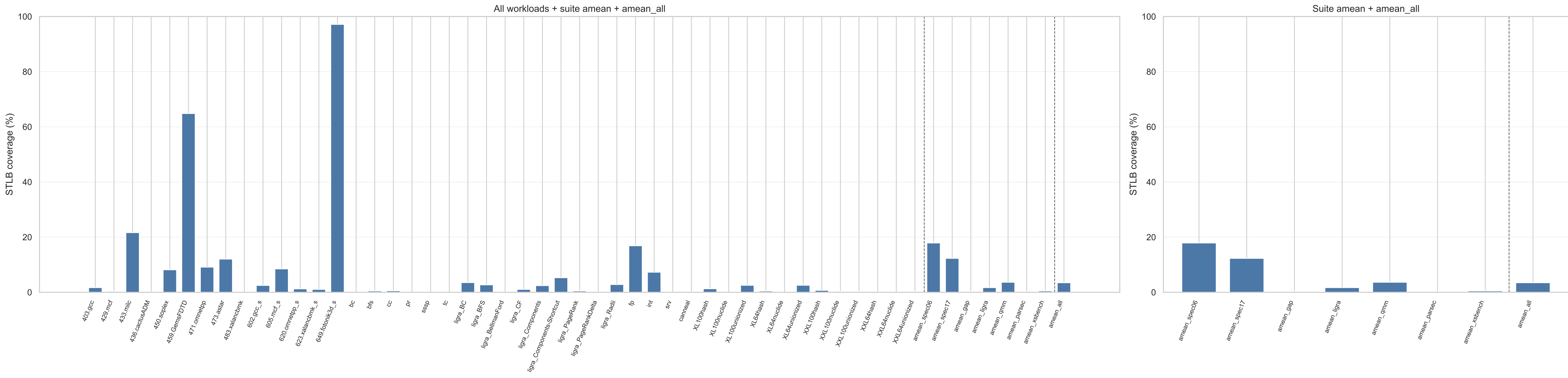
PQ drop rates: benchmark and amean_all (trace-equal)



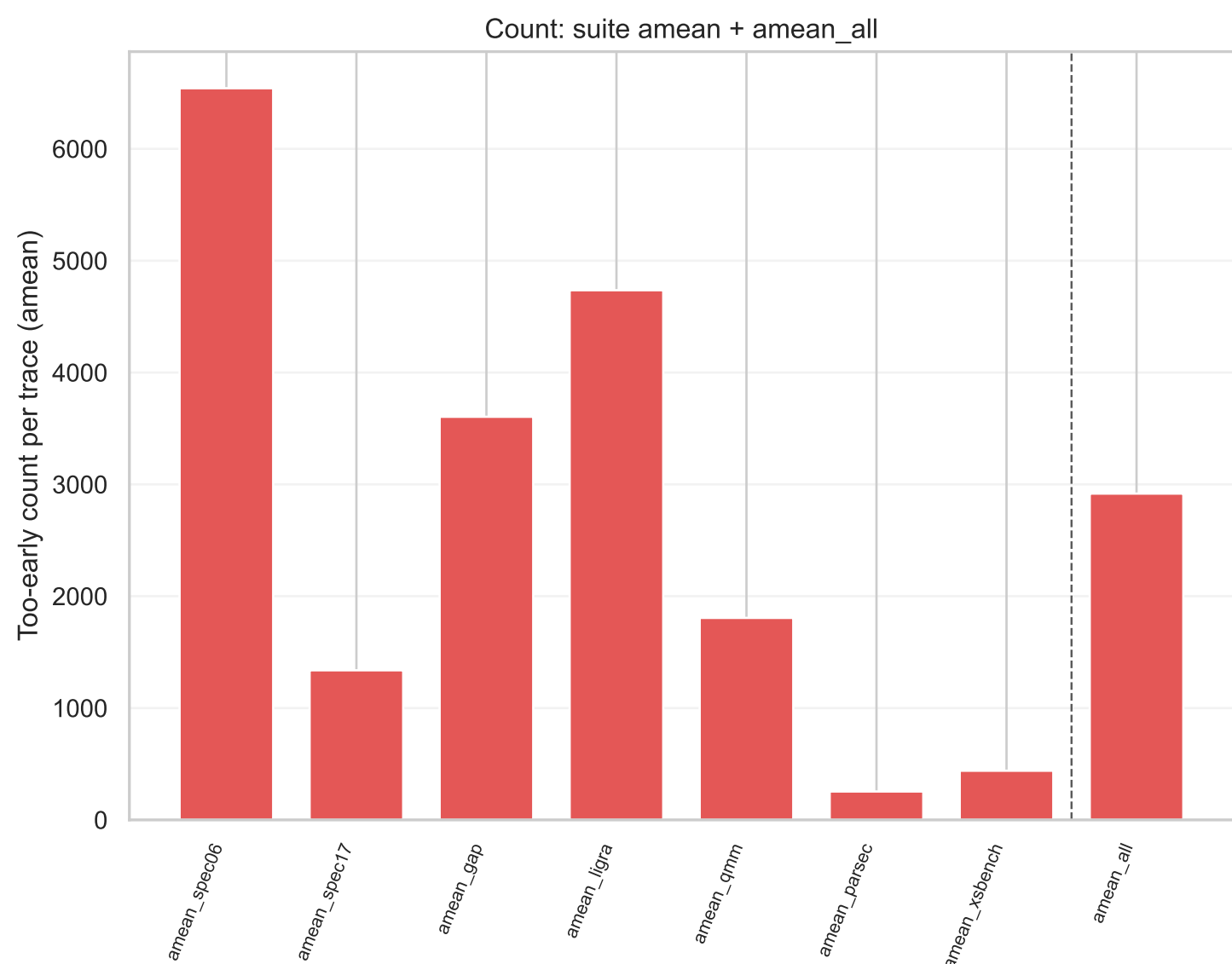
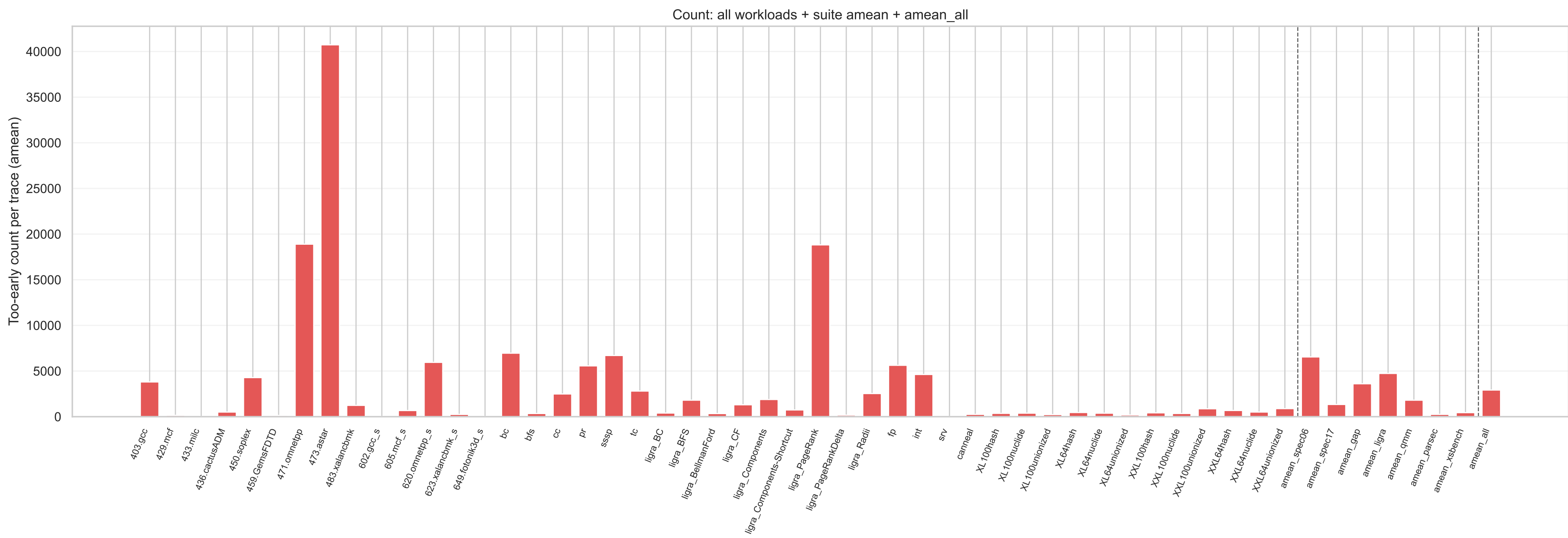
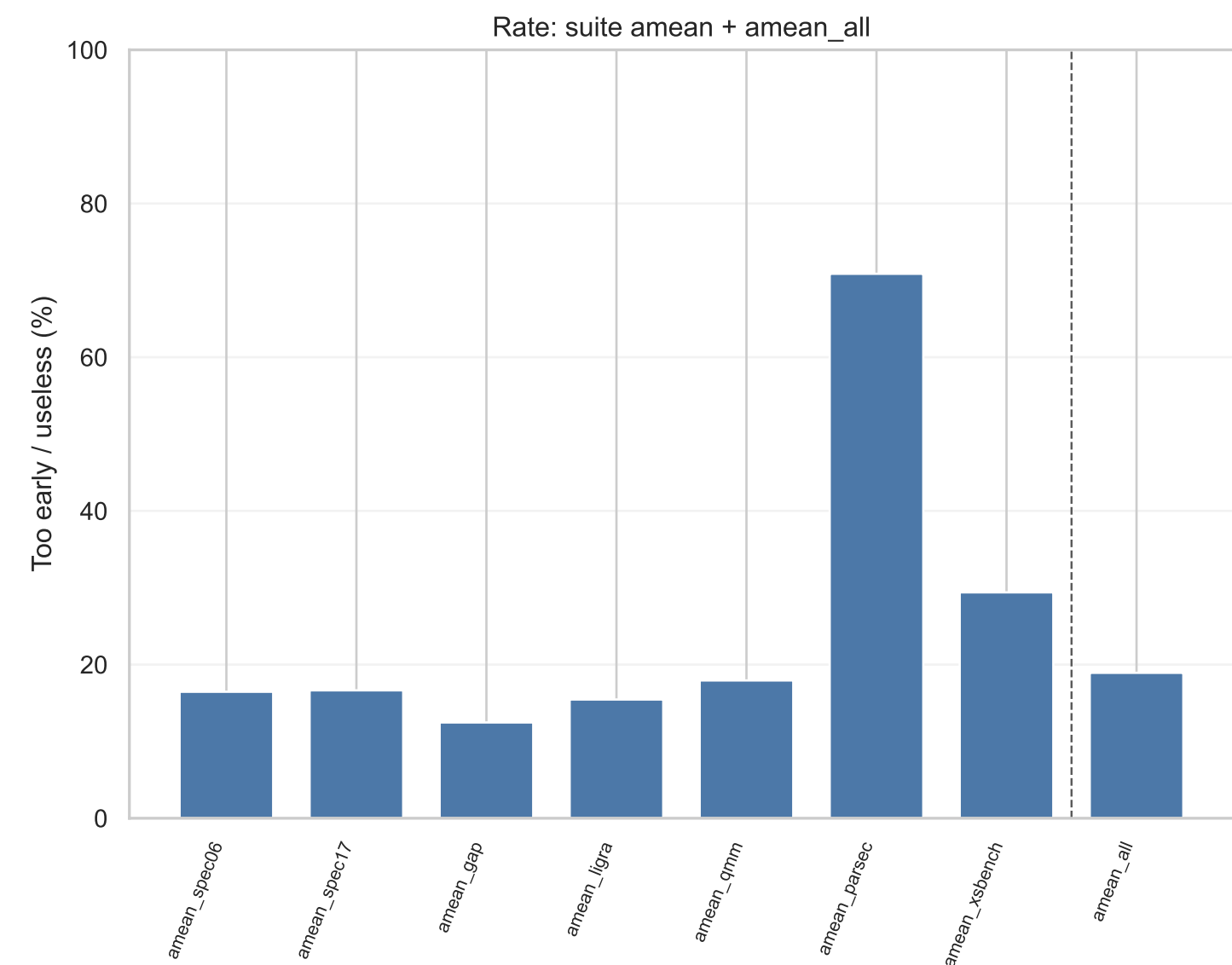
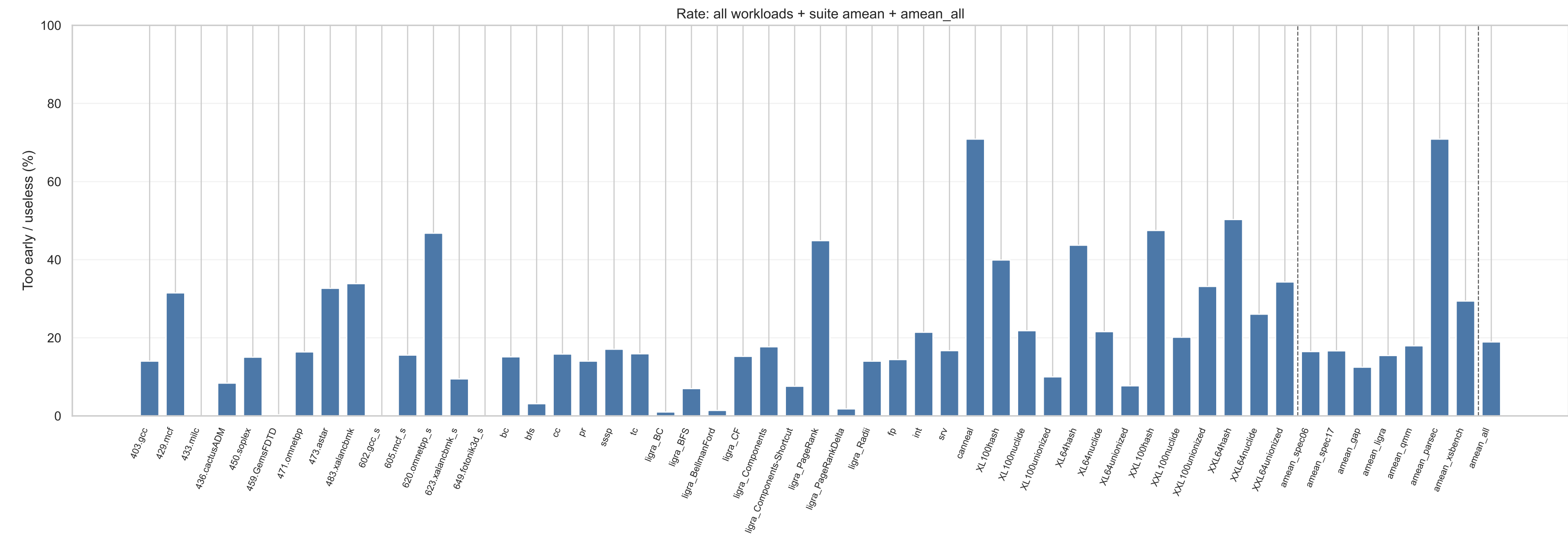
11. vBerti cross-page prefetch accuracy at STLB



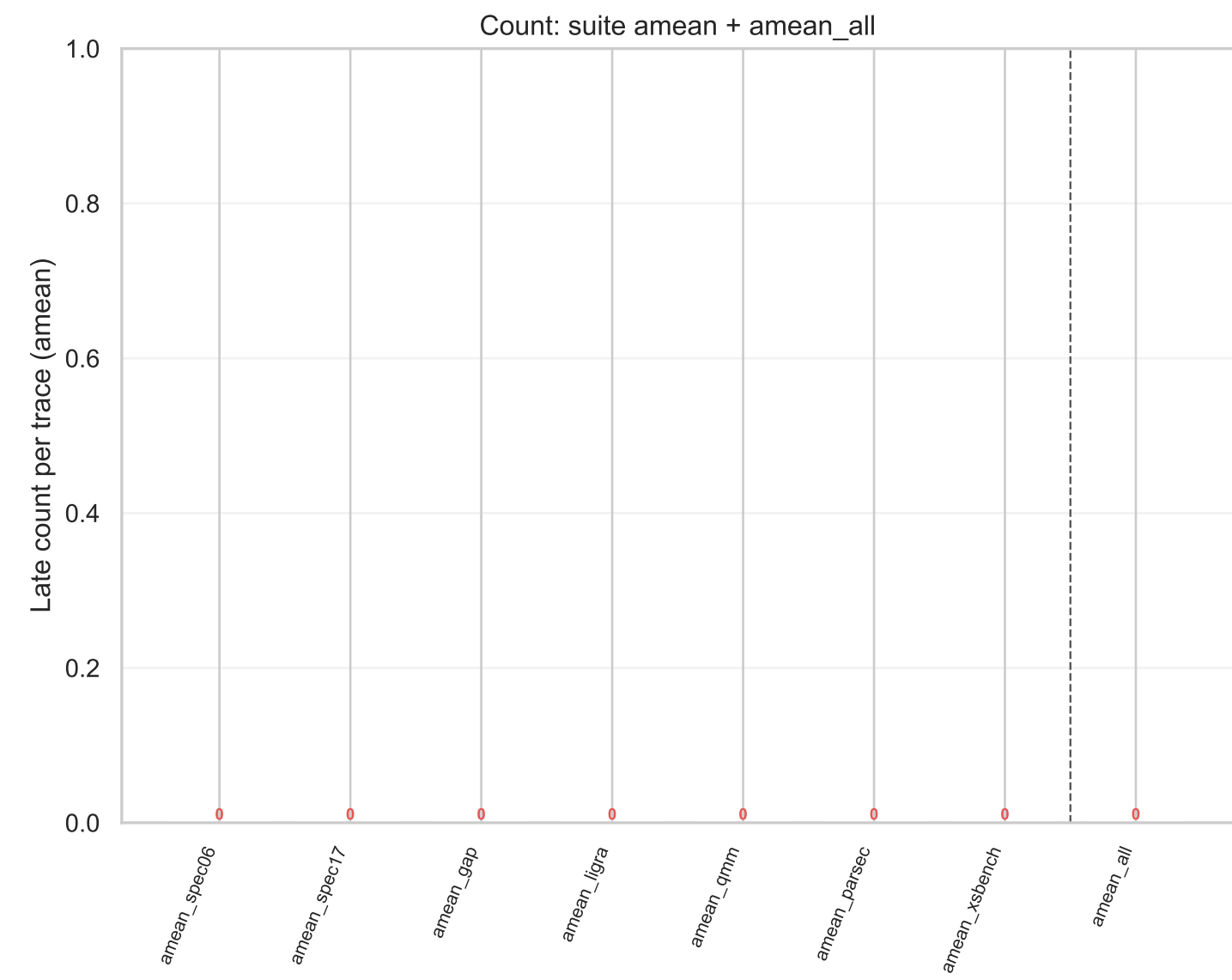
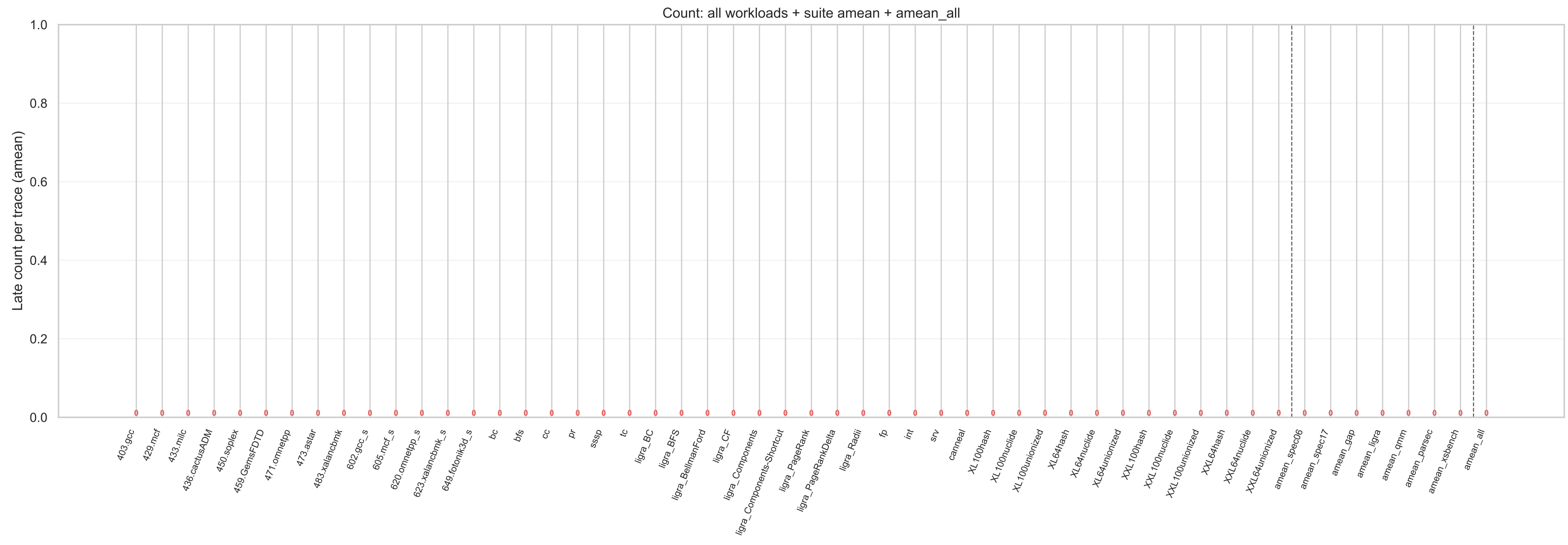
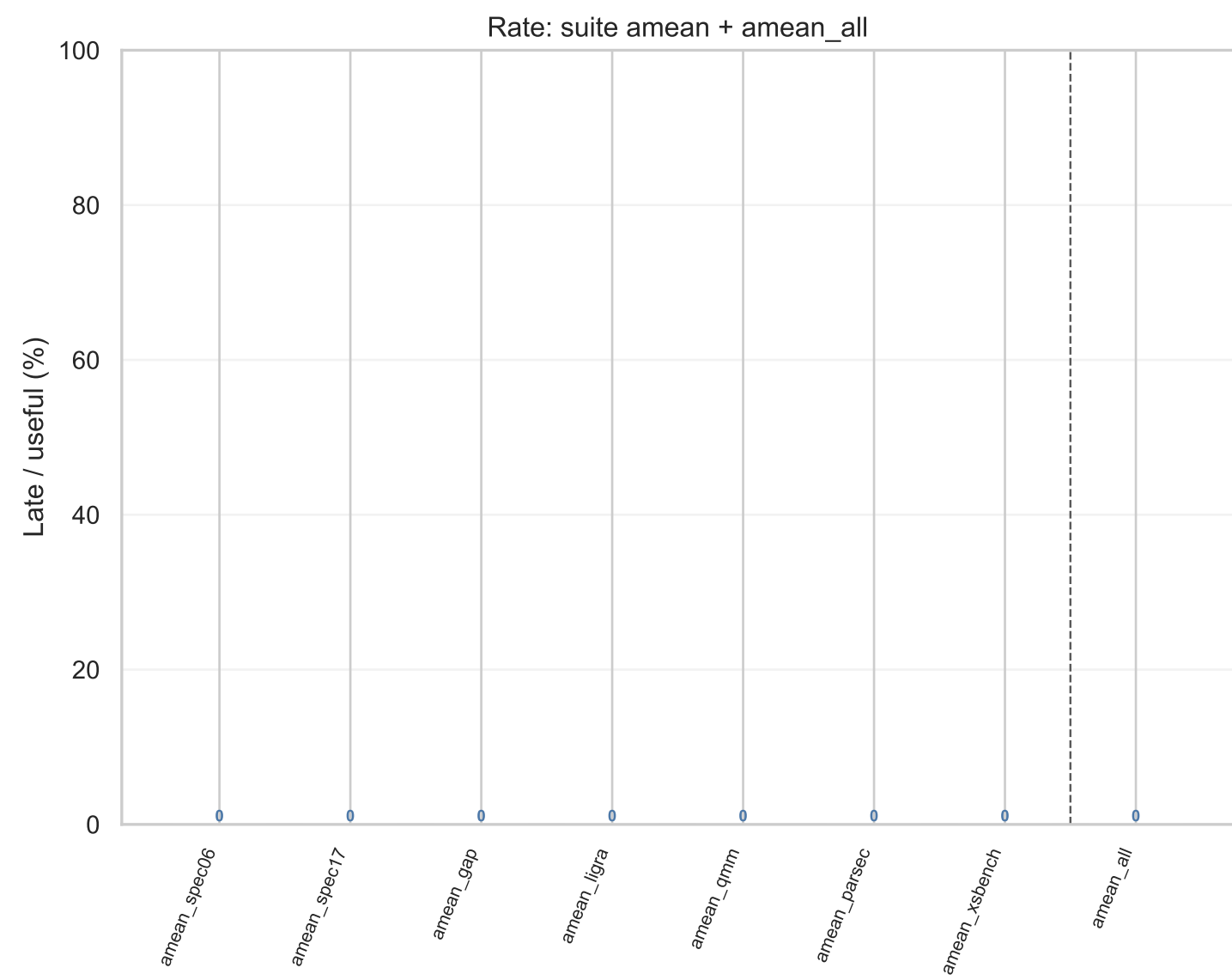
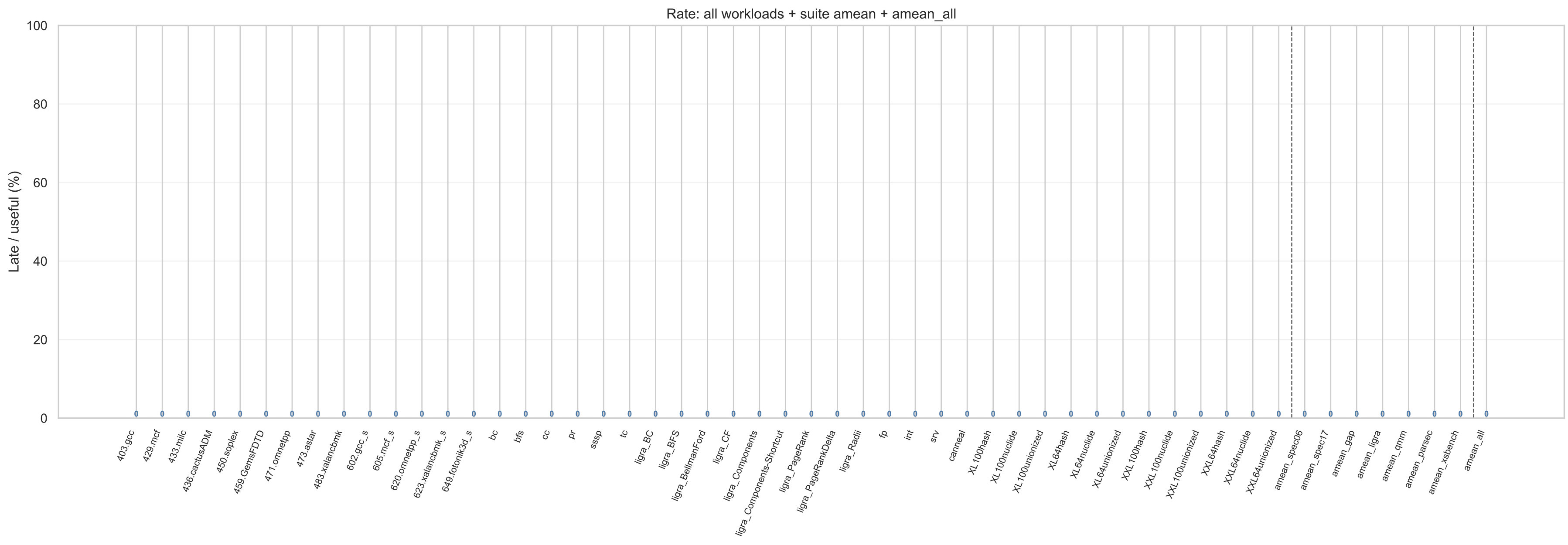
12. vBerti cross-page prefetch coverage at STLB



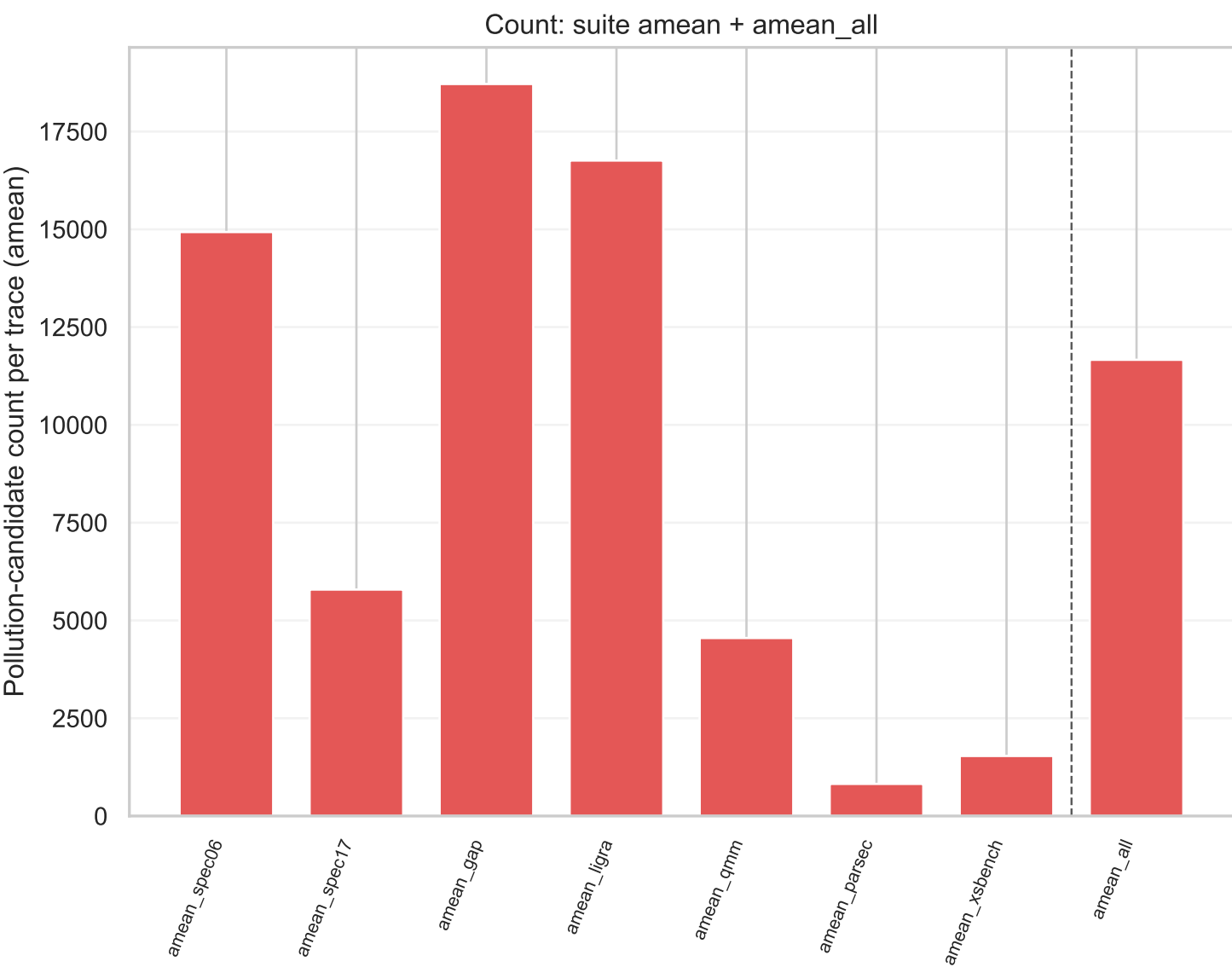
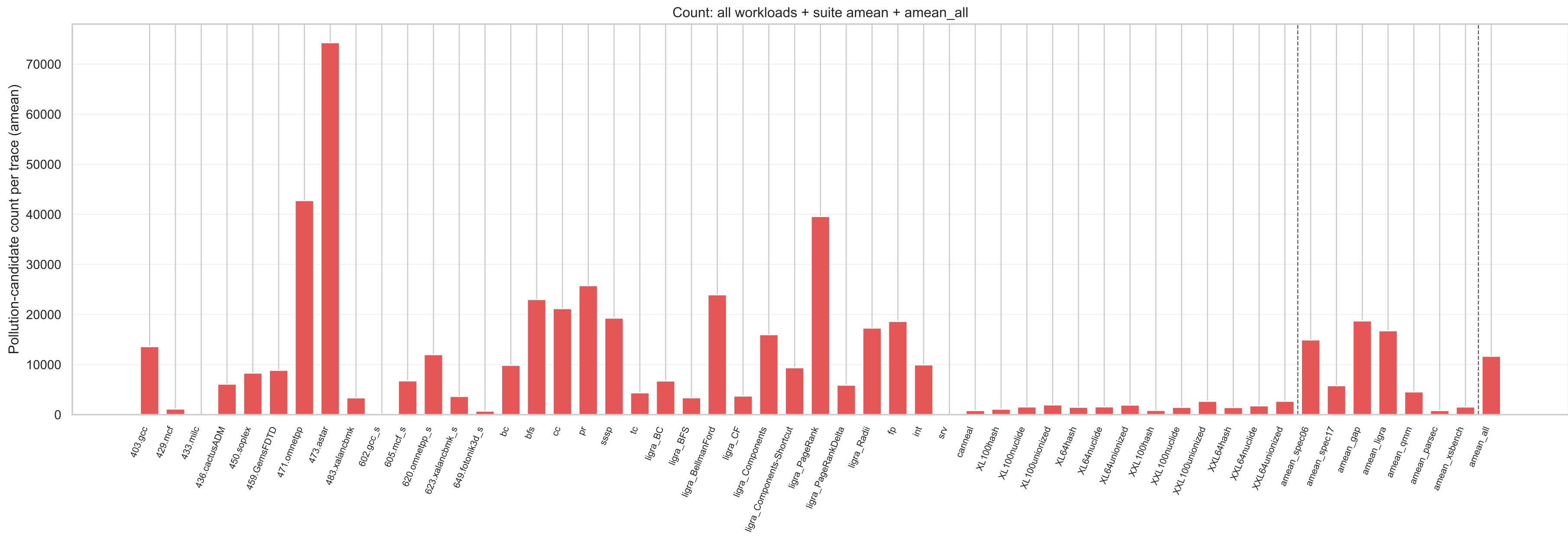
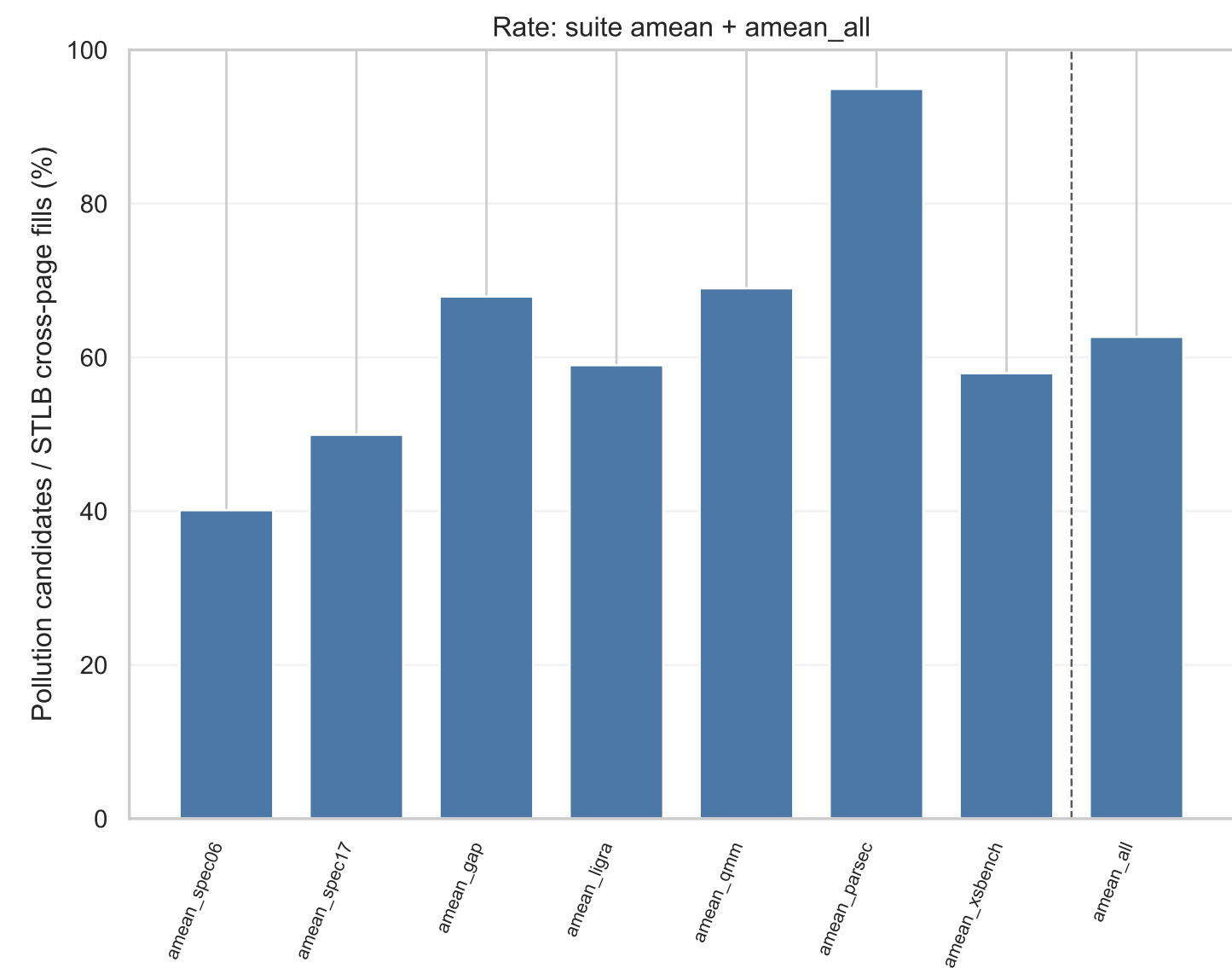
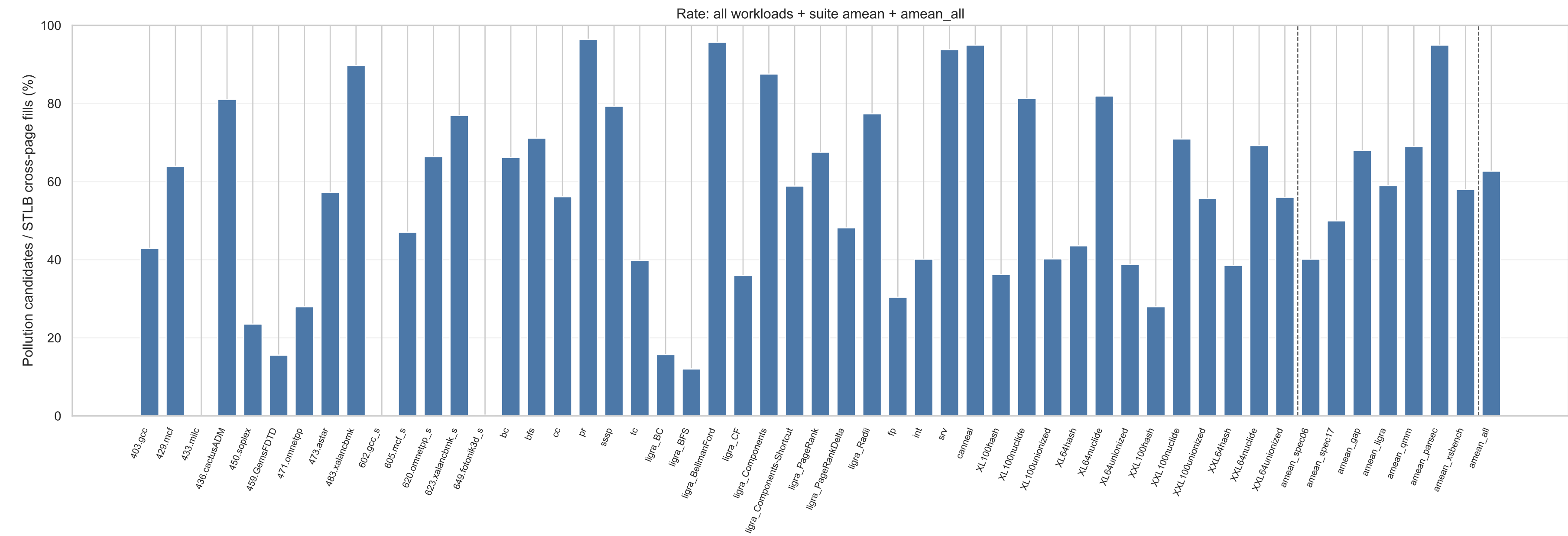
13. vBerti cross-page prefetch too-early at STLB



14. vBerti cross-page prefetch too-late at STLB



15. vBerti cross-page prefetch pollution candidates at STLB



16. vBerti end-to-end data-prefetch accuracy

